

XCS221 Assignment 7 — Logic (Extra Credit)

Due Sunday, February 2 at 11:59pm PT.

Guidelines

1. If you have a question about this homework, we encourage you to post your question on our Slack channel, at <http://xcs221-scpd.slack.com/>
2. Familiarize yourself with the collaboration and honor code policy before starting work.
3. For the coding problems, you must use the packages specified in the provided environment description. Since the autograder uses this environment, we will not be able to grade any submissions which import unexpected libraries.

Submission Instructions

Written Submission: Some questions in this assignment require a written response. For these questions, you should submit a PDF with your solutions online in the online student portal. As long as the PDF is legible and organized, the course staff has no preference between a handwritten and a typeset $\text{\LaTeX}$ submission. If you wish to typeset your submission and are new to $\text{\LaTeX}$, you can get started with the following:

- Type responses only in `submission.tex`.
- Submit the compiled PDF, **not** `submission.tex`.
- Use the commented instructions within the `Makefile` and `README.md` to get started.

Also note that your answers should be in order and clearly and correctly labeled to receive credit. Be sure to submit your final answers as a PDF and **tag all pages correctly when submitting to Gradescope**.

Coding Submission: Some questions in this assignment require a coding response. For these questions, you should submit only the `src/submission.py` file in the online student portal. For further details, see Writing Code and Running the Autograder below.

Honor code

We strongly encourage students to form study groups. Students may discuss and work on homework problems in groups. However, each student must write down the solutions independently, and without referring to written notes from the joint session. In other words, each student must understand the solution well enough in order to reconstruct it by him/herself. In addition, each student should write on the problem set the set of people with whom s/he collaborated. Further, because we occasionally reuse problem set questions from previous years, we expect students not to copy, refer to, or look at the solutions in preparing their answers. It is an honor code violation to intentionally refer to a previous year's solutions. More information regarding the Stanford honor code can be found at <https://communitystandards.stanford.edu/policies-and-guidance/honor-code>.

Writing Code and Running the Autograder

All your code should be entered into `src/submission.py`. When editing `src/submission.py`, please only make changes between the lines containing `### START_CODE_HERE ###` and `### END_CODE_HERE ###`. Do not make changes to files other than `src/submission.py`.

The unit tests in `src/grader.py` (the autograder) will be used to verify a correct submission. Run the autograder locally using the following terminal command within the `src/` subdirectory:

```
$ python grader.py
```

There are two types of unit tests used by the autograder:

- **basic:** These tests are provided to make sure that your inputs and outputs are on the right track, and that the hidden evaluation tests will be able to execute.
- **hidden:** These unit tests are the evaluated elements of the assignment, and run your code with more complex inputs and corner cases. Just because your code passed the basic local tests does not necessarily mean that they will pass all of the hidden tests. These evaluative hidden tests will be run when you submit your code to the Gradescope autograder via the online student portal, and will provide feedback on how many points you have earned.

For debugging purposes, you can run a single unit test locally. For example, you can run the test case `3a-0-basic` using the following terminal command within the `src/` subdirectory:

```
$ python grader.py 3a-0-basic
```

Before beginning this course, please walk through the [Anaconda Setup for XCS Courses](#) to familiarize yourself with the coding environment. Use the env defined in `src/environment.yml` to run your code. This is the same environment used by the online autograder.

Test Cases

The autograder is a thin wrapper over the python `unittest` framework. It can be run either locally (on your computer) or remotely (on SCPD servers). The following description demonstrates what test results will look like for both local and remote execution. For the sake of example, we will consider two generic tests: `1a-0-basic` and `1a-1-hidden`.

Local Execution - Hidden Tests

All hidden tests rely on files that are not provided to students. Therefore, the tests can only be run remotely. When a hidden test like `1a-1-hidden` is executed locally, it will produce the following result:

```
----- START 1a-1-hidden: Test multiple instances of the same word in a sentence.
----- END 1a-1-hidden [took 0:00:00.011989 (max allowed 1 seconds), ???/3 points] (hidden test ungraded)
```

Local Execution - Basic Tests

When a basic test like `1a-0-basic` passes locally, the autograder will indicate success:

```
----- START 1a-0-basic: Basic test case.
----- END 1a-0-basic [took 0:00:00.000062 (max allowed 1 seconds), 2/2 points]
```

When a basic test like `1a-0-basic` fails locally, the error is printed to the terminal, along with a stack trace indicating where the error occurred:

```
----- START 1a-0-basic: Basic test case.
<class 'AssertionError'>
{'a': 2, 'b': 1} != None ← This error caused the test to fail.
File "/Users/grinch/Local_Documents/Software/anaconda3/envs/XCS221/lib/python3.6/unittest/case.py", line 59, in testPartExecutor
    yield
File "/Users/grinch/Local_Documents/Software/anaconda3/envs/XCS221/lib/python3.6/unittest/case.py", line 605, in run
    testMethod()
File "/Users/grinch/Local_Documents/SCPD/XCS221/A1/src/graderUtil.py", line 54, in wrapper
    result = func(*args, **kwargs)
File "/Users/grinch/Local_Documents/SCPD/XCS221/A1/src/graderUtil.py", line 83, in wrapper
    result = func(*args, **kwargs)
File "/Users/grinch/Local_Documents/SCPD/XCS221/A1/src/grader.py", line 23, in test_0
    submission.extractWordFeatures("a b a") ← In this case, start your debugging
                                           in line 23 of grader.py.
File "/Users/grinch/Local_Documents/Software/anaconda3/envs/XCS221/lib/python3.6/unittest/case.py", line 829, in assertEqual
    assertion_func(first, second, msg=msg)
File "/Users/grinch/Local_Documents/Software/anaconda3/envs/XCS221/lib/python3.6/unittest/case.py", line 822, in _baseAssertEqual
    raise self.failureException(msg)
----- END 1a-0-basic [took 0:00:00.003809 (max allowed 1 seconds), 0/2 points]
```

Remote Execution

Basic and hidden tests are treated the same by the remote autograder. Here are screenshots of failed basic and hidden tests. Notice that the same information (error and stack trace) is provided as the in local autograder, now for both basic and hidden tests.

1a-0-basic) Basic test case. (0.0/2.0)

```
<class 'AssertionError': {'a': 2, 'b': 1} != None
File "/autograder/source/miniconda/envs/XCS221/lib/python3.6/unittest/case.py", line 59, in testPartExecutor
    yield
File "/autograder/source/miniconda/envs/XCS221/lib/python3.6/unittest/case.py", line 605, in run
    testMethod()
File "/autograder/source/graderUtil.py", line 54, in wrapper
    result = func(*args, **kwargs)
File "/autograder/source/graderUtil.py", line 83, in wrapper
    result = func(*args, **kwargs)
File "/autograder/source/grader.py", line 23, in test_0
    submission.extractWordFeatures("a b a"))
File "/autograder/source/miniconda/envs/XCS221/lib/python3.6/unittest/case.py", line 829, in assertEqual
    assertion_func(first, second, msg=msg)
File "/autograder/source/miniconda/envs/XCS221/lib/python3.6/unittest/case.py", line 822, in _baseAssertEqual
    raise self.failureException(msg)
```

Just like in the local autograder, this error caused the test to fail.

Just like in the local autograder, start your debugging in line 23 of grader.py.

1a-1-hidden) Test multiple instances of the same word in a sentence. (0.0/3.0)

```
<class 'AssertionError': {'a': 23, 'ab': 22, 'aa': 24, 'c': 16, 'b': 15} != None
File "/autograder/source/miniconda/envs/XCS221/lib/python3.6/unittest/case.py", line 59, in testPartExecutor
    yield
File "/autograder/source/miniconda/envs/XCS221/lib/python3.6/unittest/case.py", line 605, in run
    testMethod()
File "/autograder/source/graderUtil.py", line 54, in wrapper
    result = func(*args, **kwargs)
File "/autograder/source/graderUtil.py", line 83, in wrapper
    result = func(*args, **kwargs)
File "/autograder/source/grader.py", line 31, in test_1
    self.compare_with_solution_or_wait(submission, 'extractWordFeatures', lambda f: f(sentence))
File "/autograder/source/graderUtil.py", line 183, in compare_with_solution_or_wait
    self.assertEqual(ans1, ans2)
File "/autograder/source/miniconda/envs/XCS221/lib/python3.6/unittest/case.py", line 829, in assertEqual
    assertion_func(first, second, msg=msg)
File "/autograder/source/miniconda/envs/XCS221/lib/python3.6/unittest/case.py", line 822, in _baseAssertEqual
    raise self.failureException(msg)
```

This error caused the test to fail.

Start your debugging in line 31 of grader.py.

Finally, here is what it looks like when basic and hidden tests pass in the remote autograder.

1a-0-basic) Basic test case. (2.0/2.0)

1a-1-hidden) Test multiple instances of the same word in a sentence. (3.0/3.0)

Introduction

In this assignment, you will get some hands-on experience with logic. You'll see how logic can be used to represent the meaning of natural language sentences, and how it can be used to solve puzzles and prove theorems. Most of this assignment will be translating English into logical formulas, but in Problem 4, you will delve into the mechanics of logical inference.

To get started, launch a Python shell and try typing the following commands to add logical expressions into the knowledge base.

```
(XCS221) $ python
Python 3.6.9
Type "help", "copyright", "credits" or "license" for more information.
>>> from logic import *
>>> Rain = Atom('Rain')           # Shortcut
>>> Wet = Atom('Wet')              # Shortcut
>>> kb = createResolutionKB()      # Create the knowledge base
>>> kb.ask(Wet)
I don't know.
>>> kb.ask(Not(Wet))
I don't know.
>>> kb.tell(Implies(Rain, Wet))
I learned something.
>>> kb.ask(Wet)
I don't know.
>>> kb.tell(Rain)
I learned something.
>>> kb.tell(Wet)
I already knew that.
>>> kb.ask(Wet)
Yes.
>>> kb.ask(Not(Wet))
No.
>>> kb.tell(Not(Wet))
I don't buy that.
```

To print out the contents of the knowledge base, you can call `kb.dump()`. For the example above, you get:

```
==== Knowledge base [3 derivations] ====
* Or(Not(Rain),Wet)
* Rain
- Wet
```

In the output, ‘*’ means the fact was explicitly added by the user, and ‘-’ means that it was inferred.

Here is a table that describes how logical formulas are represented in code. Use it as a reference guide:

Name	Mathematical Notation	Code
Constant symbol	stanford	<code>Constant('stanford')</code> (must be lowercase)
Variable symbol	x	<code>Variable('\$x')</code> (must be lowercase)
Atomic formula (atom)	Rain	<code>Atom('Rain')</code> (predicate must be uppercase)
	<code>LocatedIn(stanford, x)</code>	<code>Atom('LocatedIn', 'stanford', '\$x')</code> (arguments are symbols)
Negation	$\neg \text{Rain}$	<code>Not(Atom('Rain'))</code>
Conjunction	$\text{Rain} \wedge \text{Snow}$	<code>And(Atom('Rain'), Atom('Snow'))</code>
Disjunction	$\text{Rain} \vee \text{Snow}$	<code>Or(Atom('Rain'), Atom('Snow'))</code>
Implication	$\text{Rain} \rightarrow \text{Wet}$	<code>Implies(Atom('Rain'), Atom('Wet'))</code>
Equivalence	$\text{Rain} \leftrightarrow \text{Wet}$	<code>Equiv(Atom('Rain'), Atom('Wet'))</code>
	(syntactic sugar for: $\text{Rain} \rightarrow \text{Wet} \wedge \text{Wet} \rightarrow \text{Rain}$)	
Existential quantification	$\exists x. \text{LocatedIn}(\text{stanford}, x)$	<code>Exists('\$x', Atom('LocatedIn', 'stanford', '\$x'))</code>
Universal quantification	$\forall x. \text{MadeOfAtoms}(x)$	<code>Forall('\$x', Atom('MadeOfAtoms', '\$x'))</code>

The operations `And` and `Or` only take two arguments. If we want to take a conjunction or disjunction of more than two, use `AndList` and `OrList`. For example: `AndList([Atom('A'), Atom('B'), Atom('C')])` is equivalent to `And(And(Atom('A'), Atom('B')), Atom('C'))`.

1. Propositional Logic

Write a propositional logic formula for each of the following English sentences in the given function in `submission.py`. For example, if the sentence is *"If it is raining, it is wet"*, then you would write `Implies(Atom('Rain'), Atom('Wet'))`, which would be $\text{Rain} \rightarrow \text{Wet}$ in symbols (see `examples.py`).

Note: Don't forget to return the constructed formula!

- (a) [1 point (Coding, Extra Credit)] *"If it's summer and we're in California, then it doesn't rain."*
- (b) [1 point (Coding, Extra Credit)] *"It's wet if and only if it is raining or the sprinklers are on."*
- (c) [2 points (Coding, Extra Credit)] *"Either it's day or night (but not both)."*

You can run the following command to test each formula:

```
python grader.py 1a-0-basic
```

If your formula is wrong, then the grader will provide a counterexample, which is a model that your formula and the correct formula don't agree on. For example, if you accidentally wrote `And(Atom('Rain'), Atom('Wet'))` for *"If it is raining, it is wet"*, then the grader would output the following:

```
Your formula (And(Rain,Wet)) says the following model is FALSE, but it should be TRUE:
* Rain = False
* Wet = True
* (other atoms if any) = False
```

In this model, it is not raining and it is wet, which satisfies the correct formula $\text{Rain} \rightarrow \text{Wet}$ (**TRUE**), but does not satisfy the incorrect formula $\text{Rain} \wedge \text{Wet}$ (**FALSE**). Use these counterexamples to guide you in the rest of the assignment.

2. First Order Logic

Write a first-order logic formula for each of the following English sentences in the given function in `submission.py`. For example, if the sentence is “*There is a light that shines*”, then you would write

`Exists('$x', And(Atom('Light', '$x'), Atom('Shines', '$x')))`, which would be $\exists x.\text{Light}(x) \wedge \text{Shines}(x)$ in symbols (see `examples.py`).

Tip: Python tuples can span multiple lines, which help with readability when you are writing logic expressions (some of them in this homework can get quite large)

- (a) **[0.50 points (Coding, Extra Credit)]** “*Every person has a mother.*”
- (b) **[0.50 points (Coding, Extra Credit)]** “*At least one person has no children.*”
- (c) **[0.50 points (Coding, Extra Credit)]** Create a formula which defines `Daughter(x,y)` in terms of `Female(x)` and `Child(x,y)`.
- (d) **[0.50 points (Coding, Extra Credit)]** Create a formula which defines `Grandmother(x,y)` in terms of `Female(x)` and `Parent(x,y)`.

3. Liar Puzzle

Someone crashed the server, and accusations are flying. For this problem, we will encode the evidence in first-order logic formulas to find out who crashed the server. You've narrowed it down to four suspects: John, Susan, Mark, and Nicole. You have the following information:

- John says: "It wasn't me!"
- Susan says: "It was Nicole!"
- Mark says: "No, it was Susan!"
- Nicole says: "Susan's a liar."
- You know that exactly one person is telling the truth.
- You also know exactly one person crashed the server.

- (a) **[2 points (Coding, Extra Credit)]** Fill out `liar()` to return a list of 6 formulas, one for each of the above facts. Be sure your formulas are exactly in the order specified.

You can test your code using the following commands:

```
python grader.py 3a-0-basic
python grader.py 3a-1-basic
python grader.py 3a-2-basic
python grader.py 3a-3-basic
python grader.py 3a-4-basic
python grader.py 3a-5-basic
python grader.py 3a-all-basic # Tests the conjunction of all the formulas
```

To solve the puzzle and find the answer, tell the formulas to the knowledge base and ask the query `CrashedServer('$x')`, by running:

```
python grader.py 3a-run-basic
```


4. Odd and Even Integers

In this problem, we will see how to use logic to automatically prove mathematical theorems. We will focus on encoding the theorem and leave the proving part to the logical inference algorithm. Here is the theorem:

If the following constraints hold:

1. Each number x has exactly one successor, which is not equal to x .
2. Each number is either odd or even, but not both.
3. The successor of an even number is odd.
4. The successor of an odd number is even.
5. For every number x , the successor of x is larger than x .
6. Larger is a transitive property: if x is larger than y and y is larger than z , then x is larger than z .

Then we have the following consequence:

- For each number, there is an even number larger than it.

Note: in this problem, "larger than" is just an arbitrary relation, and you should not assume it has any prior meaning. In other words, don't assume things like "a number can't be larger than itself" unless explicitly stated.

- (a) **[2 points (Coding, Extra Credit)]** Fill out `ints()` to construct six formulas for each of the constraints. The consequence has been filled out for you (`query` in the code). You can test your code using the following commands:

```
python grader.py 4a-0-basic
python grader.py 4a-1-basic
python grader.py 4a-2-basic
python grader.py 4a-3-basic
python grader.py 4a-4-basic
python grader.py 4a-5-basic
python grader.py 4a-all-basic # Tests the conjunction of all the formulas
```

To finally prove the theorem, tell the formulas to the knowledge base and ask the query by running model checking (on a finite model):

```
python grader.py 4a-run-basic
```

5. Semantic Parsing

Semantic parsing is the task of converting natural language utterances into first-order logic formulas. We have created a small set of grammar rules in the code for you in `createBaseEnglishGrammar()`. In this problem, you will add additional grammar rules to handle a wider variety of sentences. Specifically, create a `GrammarRule` for each of the following sentence structures.

- (a) [2 points (Coding, Extra Credit)] Example: *Every person likes some cat.*

General template:

```
$Clause ← every $Noun $Verb some $Noun
```

- (b) [2 points (Coding, Extra Credit)] Example: *There is some cat that every person likes.*

General template:

```
$Clause ← there is some $Noun that every $Noun $Verb
```

- (c) [2 points (Coding, Extra Credit)] Example: *If a person likes a cat then the former feeds the latter.*

General template:

```
$Clause ← if a $Noun $Verb a $Noun then the former $Verb the latter
```

After implementing these functions, you should be able to try some simple queries using `nli.py`! For example:

```
(XCS221) $ python nli.py
=====
Hello! Talk to me in English.
Tell me something new (end the sentence with '.') or ask me a question (end the sentence with '?')
.
Type 'help' for additional commands.
-----
> Every person likes some cat.

>>>> I learned something.
-----
> Every cat is a mammal.

>>>> I learned something.
-----
> Every person likes some mammal?

>>>>> Yes.
```

6. Explainability of Logic-Based Systems

The notion that AI systems should be explainable and their decisions interpretable has gained traction. For instance, [GDPR’s Article 15](#) requires that individuals be provided “[meaningful information](#)” about the logic of automated decisions. Independently of legal considerations, explainable AI protects the interests of both end users (the people employing AI to make or inform decisions) and to stakeholders (individuals affected by automated decisions). These are some of the considerations cited in the literature:

- (a) Respect: when decisions that affect you are made intelligible to you, this is considered an expression of respect for your status as a human being whose interests are worthy of consideration.
- (b) Assessing fairness of rules: knowing what rules are applied to reach a decision, we have the ability to evaluate whether the rules are fair and whether they are fairly applied to us.
- (c) Contesting and correcting decisions: understanding a decision we disagree with, allows us to contest it.
- (d) Ability to change user behavior: understanding a decision that has a negative impact on us allows us to adapt our behavior in order to obtain different results in the future.

Explainability has many different meanings (see this paper by [Andrew Selbst and Solon Barocas](#); and this paper by [Finale Doshi Velez and Been Kim](#)). Intuitively, logic-based systems should have more explainability power than machine learning-based “black-box” systems. In this problem, let us explore what kind of explainability logic-based systems might provide and which human interests are protected.

Consider the Liar’s Puzzle from Problem 3. Here, the first-order logic system takes in a set of facts and reasons over them to produce a conclusion. In this case, the conclusion judges the truthfulness of each person, which could carry with it real-world consequences. For example, someone might be fired if they are found guilty of crashing the server, whether they did it or not (AI is used widely for the allocation of healthcare resources, credit scoring, insurance, and content moderation, among other consequential domains).

(a) **[0.50 points (Written, Extra Credit)]**

In the context of a general first-order logic system, like the one used in Problem 3’s Liar Puzzle, what does the system return (besides the final answer) that could be considered an explanation?

What we expect: 1-2 sentences answering the question.

(b) **[0.50 points (Written, Extra Credit)]**

Why might this explanation not be adequate?

Hint: think about model misspecification or framing of the question posed to the logic-based system.

What we expect: 1-2 sentences answering the question.

(c) **[1 point (Written, Extra Credit)]**

Now, reflect on the four explainability considerations defined in this problem. Are any of these met by the explanations produced by a logic-based system?

What we expect: 2-4 sentences answering the question.

7. Applications of Soundness and Completeness

Two important properties in logic-based systems are soundness and completeness. A system is sound if everything that is provable (derivable) is in fact true whereas a system is complete if everything that is true has a proof (derivation).

- **Definition of Soundness:** A set of inference rules *Rules* is sound if: $\{f : KB \vdash f\} \subseteq \{f : KB \models f\}$
- **Definition of Completeness:** A set of inference rules *Rules* is complete if: $\{f : KB \vdash f\} \supseteq \{f : KB \models f\}$

(a) [1 point (Written, Extra Credit)]

GPT-3.5 is a generative LLM pre-trained on a large corpus of diverse texts from the internet. The model can be fine-tuned to perform well on a wide array of downstream tasks. However, LLMs like GPT-3.5 are prone to hallucinations - or responses that are presented as fact but actually include false or misleading information [https://en.wikipedia.org/wiki/Hallucination_\(artificial_intelligence\)](https://en.wikipedia.org/wiki/Hallucination_(artificial_intelligence)). Consider anything that's generated by the LLM to be what it derives. Applying the logic definitions of soundness and completeness, explain whether GPT-3.5 is sound and/or complete. Explain your reasoning by drawing on examples and the definitions above.

What we expect: 2-3 sentences for each definition of soundness and completeness (4-6 sentences total).

(b) [0.50 points (Written, Extra Credit)]

For this and next question, imagine you are choosing between two proof-based systems. System A is sound but not complete. System B is complete but not sound. Assume the only difference between System A and System B is the difference between soundness and completeness.

You want to deploy one of these systems in a safety-critical setting, where acting on a false conclusion has severe consequences. Which system would you choose and why?

What we expect: 2-3 sentences explaining your choice, tying in the definitions of soundness and completeness.

(c) [0.50 points (Written, Extra Credit)]

Imagine you are in a setting where you have a separate verifier system that can validate hypotheses (e.g., a human in the loop that inspects the hypotheses). You want to capture as much true information as possible in this setting. Which system would you choose and why?

What we expect: 2-3 sentences explaining your choice, tying in the definitions of soundness and completeness.

This handout includes space for every question that requires a written response. Please feel free to use it to handwrite your solutions (legibly, please). If you choose to typeset your solutions, the `README.md` for this assignment includes instructions to regenerate this handout with your typeset L^AT_EX solutions.

6.a

6.b

6.c

7.a

7.b

7.c